

The VedicMindAI Formula Cheat Sheet

20 verified Vedic Maths shortcuts from our Level 1 & Level 2 curriculum — every example checked with real arithmetic.

Level 1 — Beginner

Ekadhikena Purvena

"By one more than the previous one"

Squaring numbers ending in 5

For a number ending in 5, multiply the leading digit(s) by the next number up, then write 25 after it.

$$65^2 \rightarrow 6 \times 7 = 42, \text{ append } 25 \rightarrow 4225$$

Nikhilam Navatashcaramam Dashatah

"All from 9 and the last from 10"

Multiplying numbers near 10, 100, or 1000

Find each number's deficit from the base. Cross-subtract one deficit from the other number, then multiply the deficits for the last digits.

$$8 \times 7 \text{ (base 10)} \rightarrow \text{deficits } 2, 3 \rightarrow 8-3=5, 2 \times 3=6 \rightarrow 56$$

Nikhilam (base 100)

"All from 9 and the last from 10"

Multiplying 2-digit numbers close to 100

Same idea as above, using deficits from 100 instead of 10.

$$96 \times 97 \rightarrow \text{deficits } 4, 3 \rightarrow 96-3=93, 4 \times 3=12 \rightarrow 9312$$

Nikhilam (base 1000)

"All from 9 and the last from 10"

Multiplying 3-digit numbers close to 1000

Same idea again, using deficits from 1000.

$$998 \times 996 \rightarrow \text{deficits } 2, 4 \rightarrow 998-4=994, 2 \times 4=008 \rightarrow 994008$$

Ekadhikena Purvena (digit sums)

"By one more than the previous one"

Quick error-checking any calculation

Add a number's digits repeatedly until one digit remains — a mismatch after multiplying/adding usually means an arithmetic slip.

$$\text{digit sum of } 47 \rightarrow 4+7=11 \rightarrow 1+1=2$$

Urdhva-Tiryagbhyam

"Vertically and crosswise"

Multiplying any two 2-digit numbers

Multiply units $\times$ units, then cross-multiply and add (outer $\times$ outer + inner $\times$ inner), then tens $\times$ tens — carrying between each step.

$$23 \times 46 \rightarrow \text{units } 3 \times 6 = 18 (8, \text{carry } 1) \rightarrow \text{cross } 2 \times 6 + 3 \times 4 = 24 + 12 = 36 (6, \text{carry } 3) \rightarrow \text{tens } 2 \times 4 + 3 = 11 (1, \text{carry } 1) \rightarrow 1058$$

Urdhva-Tiryagbhyam ($\times 11$)

"Vertically and crosswise"

Multiplying any number by 11

Keep the first and last digit, and write the sum of each pair of adjacent digits in between (carrying if needed).

$$43 \times 11 \rightarrow 4, (4+3)=7, 3 \rightarrow 473$$

Nikhilam ($\times 9, 99, 999$)

"All from 9 and the last from 10"

Multiplying by 9, 99, or 999

$n \times (10^k - 1) = n \times 10^k - n$ — just shift and subtract.

$$47 \times 99 \rightarrow 4700 - 47 \rightarrow 4653$$

Level 2 — Intermediate

Paravartya Yojayet

"Transpose and apply"

Dividing by a 2-digit number close to 10 (11–19)

Use the negative of the divisor's excess over 10 as a 'flag'. Bring down the first digit, then repeatedly multiply by the flag and add the next digit.

$$196 \div 11 \rightarrow Q = 17, R = 9$$

Urdhva-Tiryagbhyam (3-digit)

"Vertically and crosswise"

Multiplying any two 3-digit numbers

Same crosswise method as 2-digit, extended to 5 column-sums (units through ten-thousands), carrying right to left.

$$123 \times 212 \rightarrow 26076$$

Urdhva-Tiryagbhyam (squaring) <i>"Vertically and crosswise"</i> Squaring any 2-digit number units ² , then 2×tens×units, then tens ² — carry right to left. $87^2 \rightarrow 49, 112, 64 \text{ with carries} \rightarrow 7569$	Anurupyena <i>"Proportionally"</i> Cubing any 2-digit number Write four terms in ratio: $a^3, 3a^2b, 3ab^2, b^3$ — then carry right to left. $23^3 \rightarrow 8, 36, 54, 27 \text{ with carries} \rightarrow 12167$
Vilokanam <i>"By mere observation"</i> Square roots of 4-digit perfect squares Pair digits from the right. The last digit tells you the root's last digit (1→1/9, 4→2/8, 9→3/7, 6→4/6, 5→5, 0→0); the leading pair tells you the first digit. $\sqrt{1600} \rightarrow 40$	Vilokanam (cube roots) <i>"By mere observation"</i> Cube roots of perfect cubes Group the last 3 digits. Cube endings are unique (0→0, 1→1, 8→2, 7→3...9→9), so the last digit is a direct lookup; the remaining group gives the first digit. $\sqrt[3]{1728} \rightarrow 12$
Navashesha (casting out nines) <i>"Remainder of nine"</i> Checking a multiplication answer Reduce both factors to a single digit sum, multiply those, and reduce again — it must match the digit sum of the real product. $23 \times 45 = 1035 \rightarrow \text{digit sums } 5 \times 0 = 0, \text{ product's digit sum } 1035 \rightarrow 0 \rightarrow \text{match, correct}$	×5 / ×25 / ×125 shortcut Multiplying by 5, 25, or 125 $\times 5 = \times 10 \div 2, \times 25 = \times 100 \div 4, \times 125 = \times 1000 \div 8.$ $84 \times 25 \rightarrow 8400 \div 4 \rightarrow 2100$
Dhvajanka (Straight Division) <i>"Flag division"</i> Dividing by any 2-digit number Split the divisor into a principal digit and a flag digit. Divide by the principal digit, subtract the flag's contribution at each step. $384 \div 21 \rightarrow Q = 18, R = 6$	Cross-multiplication Adding or subtracting fractions without finding a common denominator $a/b + c/d = (ad + bc) / bd$, then simplify. $1/2 + 1/3 \rightarrow (3+2)/6 \rightarrow 5/6$
Antyayor Dashakepi <i>"Last-digits sum to ten"</i> Multiplying 2 numbers with the same leading digit(s) and units digits that add to 10 Multiply the leading digit by one more than itself; multiply the two units digits; combine. $43 \times 47 \rightarrow 4 \times 5 = 20, 3 \times 7 = 21 \rightarrow 2021$	Calendar codes Finding the day of the week for any date Sum a year code, a century/leap adjustment, a month code, and the date — then take mod 7 to get the weekday. $\text{August } 15, 1947 \rightarrow \text{Friday}$

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